

ENVS 193DS - Spring Final

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Github Repo

My repo [here](#)

Set Up

Load packages

```
# load in packages
library (tidyverse)
library (here)
library (janitor)
library (rstatix)
library (ggeffects)
library (scales)
library (MuMIn)
library (DHARMA)
```

Load data

```
# create object for nest data
nests <- read.csv (here("data", "nest_data_final.csv"))
# create object for bike commute data
bike_commute <- read.csv(here("data", "ENVS_193DS_Personal_Data.csv"))
```

Problem 1. Research writing

a. Transparent statistical methods

In **part 1**, my coworker used a linear model (linear regression) to perform their analysis. The response variable is binary and reflects the probability that the American Avocet uses the micro habitat (1) or doesn't (0). The predictor is a continuous variable describing the distance to water edge (m).

In **part 2**, they used a chi-squared test for analysis to determine whether there is a relationship between two categorical variables. These categorical variables are bird species (American Avocet, Forster's Tern, Black-necked Stilt) and habitat type (managed pond, non-tidal marsh, salt pond, tidal marsh, tidal mudflat).

b. Suggestions for rewriting

Part 1: The data suggests that American Avocets are more likely to use micro habitats that are closer to the water's edge. We can reject our null hypothesis because the data shows a strong relationship that is unlikely due to random chance.

Part 2: The three shorebird species used the five habitat types in unequal proportions. The species showed a clear pattern of habitat preference which are unlikely to appear solely from random chance.

c. Further analysis

Another variable that could influence the probability of American Avocet micro habitat use, is Salinity (ppt) which is a continuous variable that can be measured with a salinity meter. It might have an impact on micro-habitat use because salinity influences the composition and abundance of aquatic invertebrates which serve as prey for Avocets and therefore could predict Avocet distribution.

Problem 2. Data analysis

a. Clean and wrangle

```

nests_clean <- nests |>
  # select columns of interest
  select (height_cm, case_control, ant_species) |>
  # filter to only include ant species C. mimosae, C. sjostedti, and C. nigriceps
  filter (ant_species %in% c ("AB", "RRB", "BBR")) |>
  # relevel ant species to be in the following order: 1) C. sjostedti, 2) C. mimosae, 3) C. nigriceps
  mutate (ant_species = fct_relevel (ant_species, "AB", "RRB", "BBR")) |>
  # create a new column to store the full scientific name for each ant species
  mutate (scientific_name = case_when (
    ant_species == "AB" ~ "Crematogaster sjostedti",
    ant_species == "RRB" ~ "Crematogaster mimosae",
    ant_species == "BBR" ~ "Crematogaster nigriceps"))

slice_sample(nests_clean, n = 10)

```

	height_cm	case_control	ant_species	scientific_name
1	166.0	1	RRB	Crematogaster mimosae
2	207.0	0	RRB	Crematogaster mimosae
3	163.0	0	BBR	Crematogaster nigriceps
4	130.0	0	RRB	Crematogaster mimosae
5	193.5	0	RRB	Crematogaster mimosae
6	191.0	1	RRB	Crematogaster mimosae
7	390.0	1	BBR	Crematogaster nigriceps
8	103.5	0	RRB	Crematogaster mimosae
9	163.5	0	RRB	Crematogaster mimosae
10	110.5	0	RRB	Crematogaster mimosae

b. Response variable

The response variable `case_control` is binomial and looks at whether a tree contained a nest or if it was an “available” tree. If the tree contained a nest, it received a “1”, and if the tree did not contain a nest, it received a “0”.

c. Purpose of study

- Tree height is a continuous predictor variable that is measured in cm. As tree height increases, the probability of nest occurrence most likely increases because the birds will be able to nest farther away from potential predators on the ground. **Finish this with citations**
- Acacia ant species is a categorical predictor variable with 3 levels: *Crematogaster sjostedti* (AB), *Crematogaster mimosae* (RRB), and *Crematogaster nigriceps* (BBR).

d. Table of models

Model number	Tree Height (cm)	Ant Species	Predictor list
0			no predictors (null model)
1	X	X	all predictors (full model)
2	X		Tree Height
3		X	Ant Species

e. Run the models

```
# model 0: null model
model0 <- lm (
  case_control ~ 1, # formula: no predictors
  data = nests_clean # data = clean
)

# model 1: all predictors
model1 <- lm (
  # formula: case_control as a function of tree height + ant species
  case_control ~ height_cm + ant_species,
  data = nests_clean
)

# model 2: Tree height
model2 <- lm (
  # formula: case_control as a function of tree height
  case_control ~ height_cm,
  data = nests_clean
)

# model 3: Ant species
model3 <- lm (
  # formula: case_control as a function of ant species
  case_control ~ ant_species,
  data = nests_clean
)
```

f. Select the best model

```
# calculating AIC
AICc (
  model0,
  model1,
  model2,
  model3
) |>
  # arranging output in order of AIC
  arrange (AICc)
```

	df	AICc
model11	5	263.9785
model12	3	270.7722
model13	4	295.9079
model0	2	296.3160

The best model as determined by Akaike's Information Criterion (AIC) is model 1; the full model including “case_control” as a function of all predictors (tree height and ant species). This model explains the most variation in the simplest way.

g. Check the diagnostics

h. Visualize the model predictions

i. Write a caption for your figure.

j. Calculate model predictions

k. Interpret your results

Problem 3. Affective and exploratory visualizations

a. Comparing visualizations

- My affective visualization does not include each individual observation as a data point like my exploratory data visualization does from homework 2. Instead, my affective visualization uses a bike path visual that represent the trend of the data and includes bikers overlaid to represent clusters of data points along the line.

- Both visualizations exhibit a clear positive linear shape that can be used to infer the relationship I discovered between bike commute duration and commute urgency.
- Between both visualizations, there is a clear relationship between bike commute duration and commute urgency. The less time I have before class when I leave home, the quicker I bike from home to campus.
- One piece of helpful feedback I received in workshop was that the way I represented outliers with the same object as my trendline was distracting and confusing, and I used this suggestion to create a more cohesive visualization by changing this representation. I also was recommended to better differentiate the bikers on the bottom left of my trendline to better represent that they are moving faster than other “points”. I utilized this suggestion by adding dust clouds behind the “faster” bikers.

b. Designing an analysis

One statistical analysis that I could use to answer my research question is a linear model (linear regression). Because my central question of my data collection is; “How does the amount of time I leave my house before class effect the duration of my bike commute?”, I can use a linear model to display this predicative relationship. My predictor variable is how much time I have from when I leave home until class start (minutes), and my response variable is the duration of bike commute (minutes). Both of these variables are continuous. I would expect these two variable to be connected and observe that my urgency to get to class predicts how quickly I ride my bike.

c. Check any assumptions and run your analysis

Clean data

```
# create new object for clean data
bike_commute_clean <- bike_commute |>
  # clean column names
  clean_names() |>
  # select columns of interest
  select(duration_m,
         time_before_class_min)
```

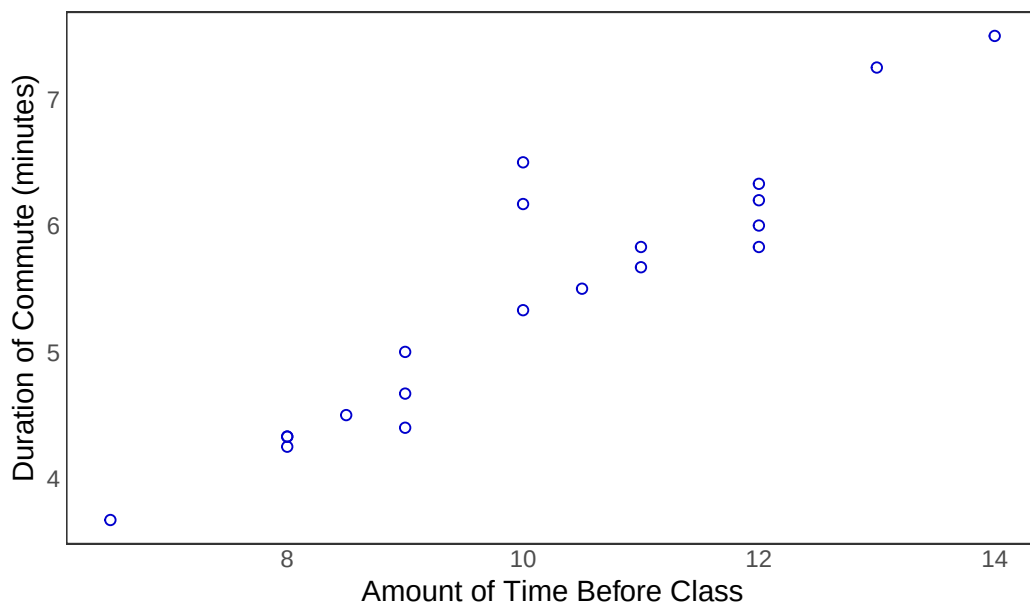
Visualize data

```

# base layer: ggplot
# data frame: bike_commute_clean
ggplot (data = bike_commute_clean,
        mapping = aes(
          # x-axis: amount of time before class
          # y-axis: duration of commute
          x = time_before_class_min,
          y = duration_m)) +
# first layer: point = scatterplot
# make data points colored, change shape
geom_point(color = "blue3",
           shape = 21) +
# Changed title, x and y-axis
labs (title = "Urgency of Classtime Bike Commute",
      x = "Amount of Time Before Class",
      y = "Duration of Commute (minutes)") +
# applied custom plot theme
theme_bw() +
# removing the panel background
theme(panel.background = element_blank()) +
# removing the panel grid
theme(panel.grid = element_blank()) +
# removing axis ticks
theme(axis.ticks = element_blank())

```


Urgency of Classtime Bike Commute



Independent observations

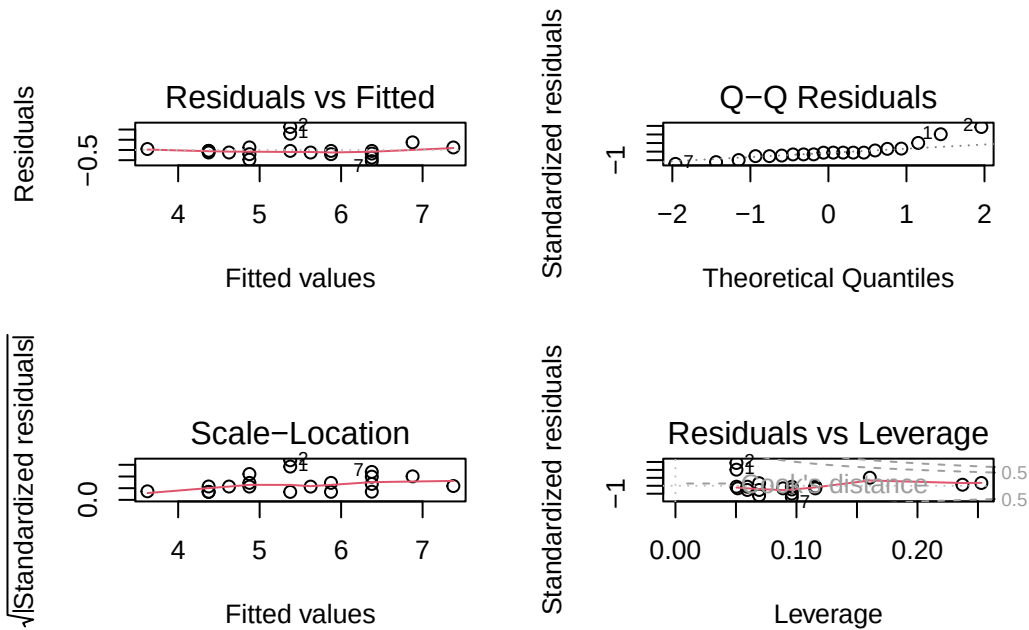
My data has independent observations. Each individual observation is independent of the observations before that. My bike commute duration to class one day, does not effect my next commute duration.

Write out model

```
# create new object for bike commute
bike_commute_model <- lm (duration_m ~ time_before_class_min, # duration as a function of time
                          data = bike_commute_clean) # data: bike_commute_clean
```

Check assumptions with diagnostic plots

```
# display diagnostic plots in 2 by 2 grid
par (mfrow = c(2, 2))
# plot bike commute model
plot (bike_commute_model)
```



Note: Using the “Residuals vs Fitted” and “Scale-Location” plots, we can determine that the residuals are evenly distributed and have constant variance (homoscedasticity). Using the “Q-Q Residuals” plot we can determine that the residuals are normally distributed. Finally, using the “Residual vs Leverage” plot, we can determine that there are no outliers that are influencing our model.

All assumptions are met

Model summary

```
#bike commute model summary
summary (bike_commute_model)
```

Call:

```
lm(formula = duration_m ~ time_before_class_min, data = bike_commute_clean)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.54666	-0.18368	-0.04634	0.06564	1.12461

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.36905	0.48971	0.754	0.461
time_before_class_min	0.50063	0.04731	10.582	3.72e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4018 on 18 degrees of freedom

Multiple R-squared: 0.8615, Adjusted R-squared: 0.8538

F-statistic: 112 on 1 and 18 DF, p-value: 3.716e-09

Note:

Intercept: 0.36905 ± 0.4897

Slope: 0.50063 ± 0.0473

Adjusted R-squared: 0.8538

P-value: $p < 0.0001$

d. Create a visualization

Model predictions

```
# new object call bike_commute_preds
bike_commute_preds <- ggpredict (bike_commute_model, # model object
                                terms = "time_before_class_min") # predictor

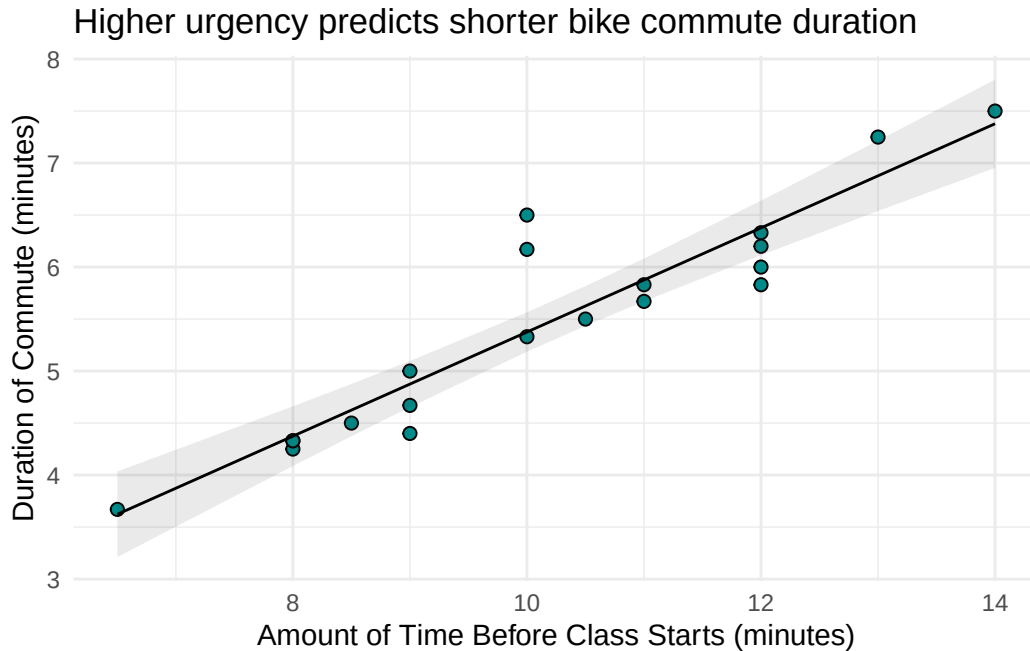
# display the predictions
bike_commute_preds
```

Predicted values of duration_m

time_before_class_min	Predicted	95% CI
6.50	3.62	3.21, 4.03
8.00	4.37	4.09, 4.66
8.50	4.62	4.37, 4.88
9.00	4.87	4.65, 5.10
10.50	5.63	5.43, 5.82
11.00	5.88	5.67, 6.08
12.00	6.38	6.11, 6.64
14.00	7.38	6.95, 7.80

Visualization

```
# base layer: ggplot
# using clean data frame
ggplot(data = bike_commute_clean,
       aes(x = time_before_class_min,
           y = duration_m)) +
# first layer: points representing individual observations
geom_point(size = 2,
           stroke = 0.5,
           fill = "cyan4",
           shape = 21) +
# second layer: ribbon representing confidence interval
# using predictions data frame
geom_ribbon(data = bike_commute_preds,
          aes(x = x,
              y = predicted,
              ymin = conf.low,
              ymax = conf.high),
          alpha = 0.1) +
# third layer: line representing model predictions
# using predictions data frame
geom_line(data = bike_commute_preds,
         aes(x = x,
             y = predicted)) +
# axis labels
labs( title = "Higher urgency predicts shorter bike commute duration",
      x = "Amount of Time Before Class Starts (minutes)",
      y = "Duration of Commute (minutes)") +
# theme
theme_minimal()
```



e. Write a caption

Figure 2: Leaving home with less time before class predicts a shorter bike commute
 Visualization displays linear relationship between the “Amount of Time Before Class Starts (minutes)” on the x-axis, and the “Duration of Commute (minutes)” on the y-axis. Blue points represent individual bike commute observations. Line represents model predictions, and ribbon displays 95% confidence interval.

f. Write about your results

My biking commute duration increased as the amount of time I had before class increased (linear regression, $F(1, 18) = 112$, $p < 0.0001$, $\alpha = 0.05$). 85% ($R^2 = 0.85$) of the variation in the model can be explained the predictor.

Based on model predictions, for every 1 minute increase in available time before class starts, I expect an increase in commute duration of 0.5 ± 0.047 (SE) minutes.

g. Sharing your affective visualization

I attended week 10 workshop! :)